

What scale buys for single-cell foundation models: a controlled probe of Geneformer V1, V2-104M, and V2-316M

William Cole

Baltimore AI, Baltimore, MD, USA

info@baltimoreai.org | ORCID [0009-0004-0578-9853](https://orcid.org/0009-0004-0578-9853)

6 May 2026

Project: Morpheus. Preprint v1, 6 May 2026.

Abstract

We probe whether Geneformer’s frozen single-cell representations encode known regulatory structure, using donor-stratified linear probes on 100,000 healthy human blood/immune cells (CEL-LxGENE Census) across three pretrained variants: V1-10M (Genecorpus-30M), V2-104M and V2-316M (both pretrained on $\sim$ 104M cells). Three probes: cell-type identity (L1), TF→target structure (L2, expression-matched negatives), and hub-TF identity (L3, top-decile out-degree in DoRothEA $\cup$ CollecTRI). All comparisons include a label-permutation selectivity control (Hewitt and Liang, 2019) and a bag-of-genes baseline. The within-V2 family forms our only clean parameter-scaling axis.

Three findings. (1) Hub-TF identity is encoded at the smallest variant (V1-10M, AUC 0.72) and exceeds the bag-of-genes baseline by approximately +0.20 AUC at every scale (V1 +0.22 [+0.13, +0.29], V2-104M +0.21 [+0.14, +0.27], V2-316M +0.20 [+0.13, +0.27]) — a gap that survives a random-initialization control at matched embedding dimensionality (random-init AUC 0.42 at 256d, 0.53 at 768d, 0.44 at 1152d), ruling out the alternative that the signal is geometric room rather than learned structure. (2) TF→target probe gains are large and selectivity-passing across V1→V2 (+0.054 AUC, $p=10^{-16}$, $n=375$ TFs) but do not improve under pure parameter scaling within V2 (V2-316M – V2-104M $\Delta=-0.003$ [−0.014, +0.008], $p=0.45$). The V1→V2 gain is therefore not separable from pretraining-corpus expansion. (3) Cell-type identity (L1) shows no measurable improvement under pure parameter scaling (V2-316M – V2-104M $\Delta=-0.007$ [−0.034, +0.017]); V2-316M fails to exceed the bag-of-genes baseline ($\Delta=+0.019$ [−0.001, +0.043], $p=0.31$; we report this as “fails to reject,” not “matches,” because we did not perform formal equivalence testing).

The contribution is methodological as much as empirical: scaling claims for single-cell foundation models that compare across model-family transitions confound parameter count with pretraining-corpus and vocabulary changes, and the only clean within-family parameter-scaling axis available to us shows no measurable gain on any probe.

1 Introduction

Single-cell foundation models (scFMs) train large transformers on tens to hundreds of millions of single-cell transcriptomes with the promise that the resulting representations will encode biological structure useful across many downstream tasks. Geneformer (Theodoris et al., 2023), scGPT (Cui et al., 2024), scFoundation (Hao et al., 2024), and Nicheformer (Schaar et al., 2025) are the canonical instances of this paradigm. Each ships a frozen embedding that practitioners use as a feature extractor for cell-type classification, batch integration, perturbation prediction, and gene-regulatory inference. The implicit claim is twofold: that pretraining at scale writes biologically meaningful structure into the parameter space, and that more pretraining — more cells, more parameters, larger vocabularies — writes more of it. Neither half of that claim has been straightforward to verify.

The empirical case for scFM scale is becoming more guarded. Boiarsky et al. (2025) benchmark six scFMs against simple baselines and find that the relative ranking is task- and tissue-dependent, with raw expression baselines competitive on several integration metrics. Kedzińska et al. (2025) report that zero-shot evaluation reveals “reliability challenges” for scGPT and Geneformer that simpler methods do not exhibit. Ahlmann-Eltze et al. (2025) show that on perturbation-effect prediction — a task scFMs were explicitly designed to address — none of the six models tested consistently beats an additive linear baseline. The pattern across these papers is not that scFMs are useless; it is that the gap between scFM embeddings and stubborn classical baselines is smaller, and more task-contingent, than the original model papers implied.

A separate question, mostly unasked in the scFM literature, is what kind of scaling produces the apparent gains. In large language models the analogous question is well posed: Kaplan et al. (2020) established that loss scales as a power-law in parameter count, dataset size, and compute over many orders of magnitude, and Hoffmann et al. (2022) — the Chinchilla paper — showed that compute-optimal training requires scaling parameters and tokens jointly, so that “more parameters at fixed data” is generically sub-optimal. The Chinchilla result reframes any pre-Chinchilla scaling claim that increased only one axis: the gain attributed to parameters was partly a gain from the data scaling that came along for the ride. The same confound is present, in starker form, in the scFM literature. The Geneformer release adds a V2 family at 104M and 316M parameters trained on a fresh ~104M-cell corpus with a new vocabulary; the V1 family at 10M parameters was trained on Genecorpus-30M with a different vocabulary. A V1-vs-V2 comparison conflates parameter count, pretraining-corpus size, and vocabulary in a single step. The only clean parameter-scaling axis the public Geneformer checkpoints offer is V2-104M $\rightarrow$ V2-316M ($3\times$ parameters, fixed corpus, fixed vocabulary). Neither the original Geneformer paper nor the recent Theodoris-group scaling paper (Venkatesh et al., 2026) isolates this axis on frozen embeddings.

Probing — training a simple supervised classifier on top of frozen representations to ask whether a target property is linearly recoverable — is the natural tool for this question. The methodology has a clean lineage in NLP: Alain and Bengio (2016) introduced linear classifier probes for intermediate-layer analysis; Conneau et al. (2018) made probing canonical for sentence embeddings; Hewitt and

Liang (2019) introduced the selectivity control (a probe that succeeds on the real task must fail on a randomized control task) that distinguishes “the property is in the embedding” from “the probe is expressive enough to memorize anything”; Belinkov (2022) reviews the failure modes that an unaudited probing study runs into. Despite this provenance, scFM evaluation rarely couples probing with the full set of controls — random-initialization, label-permutation selectivity, and a non-trivial expression-level baseline — that the NLP probing literature has settled on as a minimum.

In this paper we ask, on a single donor-stratified human blood/immune cohort of 100,000 cells, three probing questions of three Geneformer variants (V1-10M, V2-104M, V2-316M): does the frozen embedding recover (i) cell-type identity at the level of CL-ontology labels, (ii) TF→target gene-regulatory edges, and (iii) the identity of high-out-degree hub transcription factors. Each comparison is paired with a bag-of-genes baseline, a random-initialization control at matched embedding dimensionality, and a Hewitt-Liang label-permutation selectivity control. We make four contributions. First, we show that hub-TF identity is encoded in Geneformer’s input gene-token embeddings at the smallest tested variant (V1-10M, AUC 0.72) and exceeds bag-of-genes by +0.20 AUC at every scale, with a gap that survives the random-initialization control at matched dimensionality — ruling out the “geometric room” alternative. Second, we show that the large V1→V2 gain on TF→target structure (+0.054 AUC, $p=10^{-16}$) is not separable from the pretraining-corpus and vocabulary expansion that accompany the V1→V2 transition, and that the only clean parameter-scaling axis available to us (V2-104M → V2-316M) shows no measurable gain. Third, we show that on cell-type identity Geneformer’s largest publicly available variant fails to exceed the bag-of-genes baseline ($\Delta=+0.019$, $p=0.31$; we report this as “fails to reject” rather than “matches” because we do not perform formal equivalence testing). Fourth, by isolating the parameter-only axis from the corpus-expansion axis, we show that for Geneformer specifically the apparent “gains from scale” in V1-vs-V2 comparisons are concentrated in the corpus axis rather than the parameter axis — the single-cell analog of the Chinchilla observation that pure parameter scaling at fixed data is the wrong axis to push.

2 Methods

2.1 Data acquisition

We assembled a single-tissue, single-condition cell corpus from the CZ CELLxGENE Discover Census (CZI Cell Science Program et al., 2025), accessed through the cellxgene_census Python client. The Census query restricted cells to `tissue_general == "blood"`, `disease == "normal"`, `is_primary_data == True`, and one of four 10x Genomics assays (3' v2, 3' v3, 5' v1, 5' v2); restricting to a single tissue and to healthy donors removes two of the largest sources of biological variance that would otherwise confound a probe of representational structure. Donor-stratified sampling was performed in two phases: we first pulled the obs metadata only, ordered donors by random shuffle (seed 0), and admitted donors whole into the working set in order until the cumulative cell count first reached the 100,000-cell target; only the single boundary donor was randomly subsampled to bring the total to exactly 100,000 cells. This procedure preserves the

within-donor cell distribution for every donor except the boundary case and is a prerequisite for the donor-stratified cross-validation described below. The final corpus comprised 100,000 cells from 15 donors, 41 datasets, and 71 distinct cell types annotated against the Cell Ontology (Diehl et al., 2016).

Counts were normalized per cell to a target sum of 10,000 and log1p-transformed using scanpy (Wolf et al., 2018). We chose log1p(CP10K) over more elaborate transformations (e.g., scran pooling, residual transformations) on the basis of Ahlmann-Eltze and Huber (2023), who benchmarked four classes of single-cell normalization and reported that log1p followed by PCA performs as well as or better than more sophisticated alternatives across a range of downstream tasks. Geneformer’s rank-value tokenizer expects raw CP10K, so the embedding-extraction code reconstructs CP10K via `expm1` of the cached log-normalized matrix; this round trip is exact in float32. The cached corpus was stored as a sparse CSR .h5ad keyed by Ensembl gene ID, with stable per-cell identifiers carried through every downstream artifact.

The transcription-factor regulon was assembled by unioning DoRothEA confidence levels A, B, and C (Garcia-Alonso et al., 2019) with CollecTRI (Müller-Dott et al., 2023), both pulled through the decoupler Python package (Badia-i Mompel et al., 2022). After resolving HGNC symbols to Ensembl IDs through a union of the Geneformer gc30M and gc104M gene-name dictionaries (later releases winning on collision), the regulon contained 73,269 directed TF→target edges across 1,194 unique TFs. Critically, we did not apply a model-vocabulary filter to this edge file. Per-model vocabulary restriction happens at probe time through each model’s gene index, which guarantees that V1 and V2 probes evaluate against the same canonical edge set rather than against differently filtered subsets.

2.2 Donor-stratified cross-validation

We partitioned cells into five cross-validation folds at the donor level: every cell from a given donor is assigned to exactly one fold. Folds were constructed by greedy bin-packing (donors processed largest-first, each donor placed into the currently smallest fold), and a runtime assertion verified that no donor crossed fold boundaries. The audit passed for all 15 donors.

This protocol follows the principle articulated by Zimmerman et al. (2021) that cells from a common donor are not independent samples — intra-donor expression correlations exceed inter-donor correlations, producing inflated type-I error and over-optimistic generalization estimates when within-donor cells are split across train and test. Although those authors proposed mixed-effects models for differential expression rather than a cross-validation scheme, the underlying non-independence applies identically to probing. We therefore depart from the convention established in the most-cited automatic cell-identification benchmark in scRNA-seq (Abdelaal et al., 2019), which stratifies folds by cell-type proportion only and permits arbitrary donor leakage between train and test partitions. Published cell-type-classification accuracies obtained under that convention are not directly comparable to ours; our protocol is the more conservative of the two.

2.3 Model variants

We extracted frozen embeddings from three pretrained Geneformer checkpoints distributed by the original authors (Theodoris et al., 2023). V1-10M comprises approximately 10 million parameters in 6 transformer layers of width 256 and 4 attention heads per layer, with vocabulary size 25,424 (the Genecorpus-30M, or “gc30M”, dictionary). It was pretrained on Genecorpus-30M (~30 million cells). V2-104M comprises approximately 104 million parameters in 12 layers of width 768 with 12 heads, with vocabulary size 20,275 (the gc104M dictionary), pretrained on a refreshed corpus of approximately 104 million cells. V2-316M comprises approximately 316 million parameters in 18 layers of width 1,152 with 18 heads, sharing both the gc104M vocabulary and the V2 pretraining corpus.

Three changes co-occur across the V1→V2 transition: parameter count ($10\times$ larger), pretraining-corpus size ($\sim 3.5\times$ larger), and vocabulary (gc30M → gc104M, with associated changes in tokenizer median-expression statistics). The V1→V2 step is therefore confounded as a parameter-scaling experiment. The only clean parameter-scaling axis available across publicly released Geneformer checkpoints is V2-104M → V2-316M, which holds pretraining corpus and vocabulary fixed and triples parameter count. Every cross-variant comparison in this paper is labelled explicitly as either “confounded (V1→V2)” or “parameter-only (within V2)”.

2.4 Embedding extraction

Cell embeddings were computed as the mean of the final-layer hidden state across non-padding tokens, following the empirical finding by Reimers and Gurevych (2019) that mean-pooling over non-pad tokens yields the strongest fixed-size sentence representation for downstream linear probes among standard pooling schemes (CLS, max, mean). Inference used transformers (Wolf et al., 2020) and torch (Paszke et al., 2019) with weights frozen and gradients disabled. Tokenization followed the published Geneformer rank-value scheme: per-cell expression was median-scaled using each variant’s gene-median dictionary, sorted in descending order, truncated to the top 2,048 tokens, and dynamically truncated within each batch to the longest non-pad sequence. The pipeline streamed cells.h5ad in chunks of 2,000 cells, densifying only the active chunk and writing pooled embeddings into a numpy memmap; the full expression matrix is never materialized in dense form, keeping peak memory below the 24 GB target on the laptop runs.

Gene embeddings were taken directly from the model’s input token-embedding lookup table (model.embeddings.word_embeddings.weight) rather than from any contextualized aggregate. This isolates what pretraining writes into the parameter space without per-batch confounds — the same gene token receives the same vector regardless of its co-occurring genes — and is the appropriate representation for the gene-relational L2 and L3 probes, where the question is whether the model has organized its gene vocabulary in a way that respects regulatory relationships.

2.5 Controls

Three controls were run at the embedding-extraction or probe stage; a fourth (the donor-leakage audit) was described above. First, random-initialization controls were extracted at V1-10M and V2-104M dimensionalities by instantiating each architecture from its config without loading pretrained weights and running the identical extraction pipeline; this baseline isolates contributions of the architecture and the input vocabulary from contributions of pretraining (Zhang and Bowman, 2018). Second, the bag-of-genes baseline uses, as a cell embedding, the vocabulary-restricted $\log_{10}(\text{CP10K})$ vector stored as a 22,029-dimensional sparse vector, and as a gene embedding, the leading 256 components of a TruncatedSVD decomposition of the gene-by-cell matrix (matching V1’s 256-dimensional embedding). This baseline answers whether linear-probe-accessible information beyond what is already linear in raw expression is being added by the foundation model. Third, label-permutation selectivity (Hewitt and Liang, 2019) reruns the L2 probe with each TF’s positive target set replaced by a random gene set of the same cardinality drawn from the model’s vocabulary; a probe that does not collapse on this perturbation is overfitting to embedding capacity rather than to genuine TF→target structure.

2.6 Probes

All primary probes used scikit-learn LogisticRegression with $C = 1.0$, the conservative probe class recommended by Hewitt and Liang (2019) and Belinkov (2022). Sensitivity to probe capacity is reported separately below.

The Layer 1 probe (cell-type identity) is a multinomial logistic regression on cell embeddings, evaluated by macro-F1 across the donor-stratified 5-fold splits. Cell types with fewer than 200 cells were dropped to keep folds populated; 39 of the 71 Cell Ontology types in the corpus passed this threshold, retaining 98,434 of 100,000 cells. Features were standardized (mean-free for sparse inputs).

The Layer 2 probe (TF→target identity) trains, per transcription factor, a binary logistic regression that distinguishes that TF’s targets from a matched set of non-target genes, on gene embeddings. We restricted to TFs with at least 30 in-vocabulary targets, leaving 375 of 1,194 TFs. Negatives were drawn under expression-matched sampling: each gene was binned into one of ten quantiles of mean log-expression computed over cells.h5ad, and negatives were sampled to match the bin distribution of that TF’s positive set. This controls for the obvious confound that highly expressed genes are more easily separated by any embedding-based probe, regardless of whether the relevant signal is regulatory; it is a deliberate departure from the random-pair convention used in BEELINE (Pratapa et al., 2020). Within each TF, a random 80/20 gene split (seed 0) was used as train/test, and AUC was reported per TF.

The Layer 3 probe (hub identity) tests whether top-decile out-degree TFs are linearly separable from peripheral TFs on gene embeddings. Out-degree was computed on the unioned regulon, restricted to TFs whose Ensembl IDs lie in the model vocabulary; the top-decile threshold corresponded to ≥ 143 targets, identifying 120 of 1,194 TFs as hubs. We assessed this binary classifier by 5-fold

cross-validation across TFs (folds drawn by uniform integer assignment), reporting AUC per fold.

Beyond the supervised probes, we computed a representational similarity analysis (Kriegeskorte et al., 2008): the Spearman correlation between pairwise cosine similarity of TF gene embeddings and the binary, symmetrized TF-by-TF adjacency matrix from the unioned regulon, computed on the upper triangle over the 1,194-TF subset.

2.7 Sensitivity sweep

To anticipate reviewer concerns about specific methodological choices, we re-ran the V1, V2-104M, and V2-316M probes under three perturbations of the default protocol. (a) Probe class: a one-hidden-layer MLP with 64 hidden units (MLPClassifier, default sklearn settings, early stopping on) replaced the linear probe. (b) Negative sampling for L2: random-pair negatives drawn uniformly from non-targets replaced the expression-matched scheme. (c) Regulon source: each of DoRothEA-only and CollecTRI-only restricted views replaced the union, identifying which observations are robust across the two regulon resources and which are resource-specific.

2.8 Statistics

All paired comparisons used the Wilcoxon signed-rank test (Wilcoxon, 1945) on per-fold or per-TF differences, with bootstrap 95% confidence intervals on the mean difference (Efron, 1979) computed in parallel (2,000 resamples, seed = 0). For the L1 and L3 probes, pairs are CV folds ($n = 5$); for L2, pairs are TFs ($n = 375$ under default settings, fewer under regulon-restricted sensitivities). With $n = 5$ paired observations, the smallest two-sided Wilcoxon p-value attainable is 0.0625, which we report unmodified — an L1/L3 within-V2 null at $p = 0.0625$ is therefore at the power floor of the test, and effect-size confidence intervals are the primary inferential statistic for those comparisons. We make this caveat explicit in every fold-level table rather than relying on the reader to remember the $n = 5$ floor.

2.9 Compute

V1-10M extraction (cell embeddings, gene embeddings, attention sample) ran end-to-end on a 24 GB MacBook Air (Apple M3) using the PyTorch MPS backend in approximately 3 hours. V2-104M and V2-316M, both larger than fits comfortably on consumer Apple Silicon for the full 100,000-cell pass, were run on AWS EC2 g5.xlarge instances (NVIDIA A10G, 24 GB VRAM) in 1 hour 17 minutes (batch size 64) and 3 hours 24 minutes (batch size 32), respectively. Each cloud run cost approximately five US dollars at on-demand pricing. All probe and analysis code ran on the laptop. Software versions are pinned via uv.lock; the deterministic seed (0) is threaded through every random component (donor ordering, fold construction, negative sampling, train/test splits, bootstrap resampling, classifier initialization).

3 Results

We probed three pretrained Geneformer variants — V1 (~10M parameters, 256-d embeddings, ~30M-cell pretraining corpus), V2-104M (768-d, ~104M-cell pretraining corpus), and V2-316M (1152-d, ~104M-cell pretraining corpus) — with three layered probes against four controls (bag-of-genes, random-init at matched dimensionality, label permutation, and a leakage audit). All probes used frozen embeddings, donor-stratified five-fold splits, and identical preprocessing across variants. The L1 (cell-type identity) probe is scored across five donor-stratified folds ($n=5$); the L3 (hub identity) probe is scored across five TF-stratified folds ($n=5$; TF assignment to folds is uniform random per Methods); the L2 (TF→target edges) probe is scored per TF, yielding $n=375$ paired observations after vocabulary filtering. The full per-comparison statistics are reported in Supplementary Table S1; we summarise the four headline contrasts (V1→V2-104M, V1→V2-316M, within-V2, V2-316M — bag-of-genes) below and visualise them in Figures 1–4.

3.1 Hub-TF identity is encoded already at the smallest variant

The L3 hub-identity probe asks whether a logistic-regression classifier can discriminate the top-decile out-degree TFs in the post-vocab unioned regulon ($\text{DoRothEA} \cup \text{CollecTRI}$) from non-hub genes using the gene’s own embedding. All three trained Geneformer variants supported this read-out substantially above chance. Mean fold-level AUC was 0.72 for V1, 0.71 for V2-104M, and 0.70 for V2-316M (Figure 1C); 95% CIs across folds were [0.69, 0.75], [0.70, 0.72], and [0.69, 0.71] respectively. The bag-of-genes baseline performed at chance (mean AUC 0.499 [0.434, 0.568]), and three random-init Geneformer transformers — instantiated at the same architecture and dimensionality as their trained counterparts but never exposed to expression data — likewise failed to encode hub identity (V1-rand 0.42 [0.39, 0.46]; V2-104M-rand 0.53 [0.49, 0.57]; V2-316M-rand 0.44 [0.41, 0.48]; Figure 1C, Figure 3).

The trained-versus-random-init gap is the geometry control we care most about, because it rules out the trivial explanation that hub TFs are simply embedded into a higher-dimensional, easier-to-shatter subspace by the transformer architecture per se. At every dimensionality the trained model beat its random-init twin by a wide margin (paired bootstrap, 2,000 resamples, seed = 0): $\Delta = +0.294$ [+0.242, +0.364] at 256-d (V1), $\Delta = +0.179$ [+0.143, +0.215] at 768-d (V2-104M), and $\Delta = +0.256$ [+0.209, +0.297] at 1152-d (V2-316M). All three trained-vs-random-init contrasts hit the $n=5$ Wilcoxon floor of $p = 0.0625$ with the same sign. Because the three random initializations are independent draws, treating the three matched-dimensionality contrasts as independent Bernoulli trials yields a joint probability of $(1/2)^3 = 1/8$ under H_0 (one-sided sign-test). The trained-vs-bag-of-genes contrasts were similarly all positive at every scale (V1 – bag $\Delta = +0.22$ [+0.13, +0.29], V2-104M – bag $\Delta = +0.21$ [+0.14, +0.27], V2-316M – bag $\Delta = +0.20$ [+0.13, +0.27], each at $p = 0.0625$), but these three contrasts share the bag-of-genes baseline under H_0 and additionally share trained-AUC components with their corresponding trained-vs-random contrasts; we therefore do not combine them with the random-init sign-test into a single joint p -value, since the independence assumption such a combination would require is not supportable from the data. Critically, scaling the corpus and parameter count from V1 to V2-316M did not improve hub-identity readout: V2-

316M – V1 $\Delta = -0.020$ $[-0.052, +0.012]$, $p = 0.625$, and V2-316M – V2-104M $\Delta = -0.011$ $[-0.028, +0.007]$, $p = 0.44$ (Figure 2). Hub-TF identity, in other words, is fully captured at V1 scale and neither lost nor improved by $31\times$ more parameters (V1→V2-316M) and $\sim 3.5\times$ more pretraining cells.

3.2 L2 TF→target edge readout gains from V1→V2 do not survive within-V2

Layer 2 asks the harder question — given a TF embedding and a candidate target embedding, can a logistic regression decide whether the directed regulatory edge exists in DoRothEA (or, where DoRothEA is too sparse, in CollecTRI)? Each of the 375 retained TFs gives one AUC per fold, scored against degree-matched negatives drawn from the same in-vocab pool. Median per-TF AUCs were 0.698 for V1, 0.761 for V2-104M, and 0.755 for V2-316M (Figure 1B). Bag-of-genes (median 0.581) and label-permuted Geneformer (median 0.500 across all three scales) confirmed that this signal is genuine and not a co-occurrence artefact.

The V1→V2-104M jump was large and tightly estimated: paired $\Delta = +0.054$ $[+0.041, +0.068]$, Wilcoxon $p = 2.4 \times 10^{-16}$ ($n = 375$). V1→V2-316M was almost identical in magnitude ($\Delta = +0.051$ $[+0.038, +0.066]$, $p = 3.2 \times 10^{-15}$). However the within-V2 contrast was a clean null: V2-316M – V2-104M $\Delta = -0.003$ $[-0.014, +0.008]$, Wilcoxon $p = 0.45$ ($n = 375$). The $\sim 3\times$ jump in parameter count from 104M to 316M, holding the pretraining corpus fixed, produced no detectable improvement on TF→target readout. Both V2 variants did, however, beat bag-of-genes by a wide margin (V2-316M – bag $\Delta = +0.167$ $[+0.151, +0.182]$, $p = 4 \times 10^{-50}$; V2-104M – bag indistinguishable from V2-316M – bag), so the L2 effect is real — it simply does not scale within V2 (Figure 2). The most parsimonious reading is that the V1→V2 step buys most of its L2 gain from the corpus expansion ($\sim 30\text{M} \rightarrow \sim 104\text{M}$ cells) and the gc30M→gc104M vocabulary change between Geneformer V1 and V2, not from raw parameter count.

3.3 L1 cell-type identity does not exceed bag-of-genes at any scale

The L1 probe trains an L2-regularised multinomial logistic regression to predict the donor-stratified cell-type label (39 cell types passing the ≥ 200 -cell threshold, of 71 in the full corpus) from the cell embedding. Mean macro-F1 per scale was 0.354 $[0.291, 0.412]$ for V1, 0.430 $[0.340, 0.520]$ for V2-104M, and 0.424 $[0.342, 0.505]$ for V2-316M; the bag-of-genes baseline was 0.404 $[0.339, 0.465]$ (Figure 1A). Neither V2 variant exceeded bag-of-genes within fold-level uncertainty: V2-316M – bag-of-genes paired $\Delta = +0.019$ $[-0.001, +0.043]$, Wilcoxon $p = 0.31$. We fail to reject the null that V2-316M cell-type readout matches bag-of-genes at this scale and fold count ($n = 5$). The within-V2 contrast was likewise null: V2-316M – V2-104M $\Delta = -0.007$ $[-0.034, +0.017]$, Wilcoxon $p = 1.00$. Random-init transformers all performed near chance (V1-rand 0.219; V2-104M-rand 0.191; V2-316M-rand 0.149), confirming that the bag-of-genes baseline — not the random-init baseline — is the relevant headline comparator for L1.

The fact that V2 did not beat a count-vector baseline on cell-type identity is striking, and we considered the possibility that the final-layer extraction we used was suboptimal. To address this we ran a layer-wise probe on V2-104M, training a fresh L1 classifier on the gene-pooled embedding

from each of the 13 transformer layers (layer 0 = token embedding, layer 12 = final block; Figure 4). Mean macro-F1 ranged from 0.393 (layer 0) to 0.430 (layer 12); the final layer (layer 12) was the strongest at 0.430, with layer 9 tied to three decimal places (0.430) and no other layer within 0.005. No middle layer crossed the bag-of-genes line. The L1 null is therefore not an artefact of extracting from the wrong layer — Geneformer V2’s cell-type-discriminative information is, at best, no better than its input gene tokens for this 39-way immune-typing task on this corpus.

3.4 Representational-similarity is null at every scale

To complement the supervised probes we computed the Spearman correlation between each model’s pairwise gene-embedding similarity matrix and the symmetric DoRothEA TF→target adjacency, restricted to in-vocab TF–TF pairs ($n = 712,221$ pairs per matrix). All three trained variants returned essentially zero RSA: V1 $\rho = -0.028$, V2-104M $\rho = -0.004$, V2-316M $\rho = +0.022$. Random-init transformers were similar (-0.003 to $+0.001$ -0.003), as was bag-of-genes ($\rho = +0.026$). No scale produced an RSA signal that distinguished trained from untrained models. We read this as evidence that the regulatory information that the supervised probes recover from L2 and L3 is not arranged in the gene-by-gene cosine geometry — it is decodable, but not isotropic.

3.5 Sensitivity analyses confirm the headline pattern

Three pre-registered sensitivity analyses were run on every probe×model combination to stress-test the headline claims.

Probe family. Replacing the linear probe with a small MLP (one hidden layer of 64 units; sklearn defaults with early stopping) reduced rather than improved every L2 and L3 contrast: V1 L2 MLP median AUC was 0.587 (down from 0.698 linear); V2-316M L2 MLP 0.701 (down from 0.755); V2-316M L3 MLP mean 0.567 (down from V2-316M trained mean 0.700). The drop is consistent with overfitting on small per-TF training sets and confirms that linear separability — the property we care about for “is information present in the embedding” — is a more reliable readout here than expressive non-linear probes. The L1 cell-type ranking under MLP was unchanged (V2-104M MLP 0.461, V2-316M MLP 0.460, bag MLP 0.366, V1 MLP 0.355), so the L1 V2 $\approx$ bag-of-genes story does not flip under a more expressive probe.

Negative-sampling scheme for L2. Switching from expression-matched negatives to random-pair negatives raised every per-TF AUC (e.g. V2-316M median 0.781 vs 0.755 matched), as expected when the negatives become easier. The V1→V2 ordering held but the gap shrank under the easier task: V2-316M – V1 paired $\Delta = +0.015$ [$+0.003$, $+0.027$], $p = 0.011$ under random negatives, compared with $+0.051$ [$+0.038$, $+0.066$] under matched. V2-316M – V2-104M remained null under both schemes. Random-pair negatives compress the dynamic range of the probe (highly expressed targets are easy regardless of regulatory signal), so the smaller V1→V2 gap under random negatives is exactly the direction expected; the qualitative ordering — V1→V2 jump real, within-V2 jump absent — is preserved.

Regulon source. Restricting L2 to the higher-confidence DoRothEA-only subset (214 TFs) and re-

running with the broader CollecTRI prior (304 TFs) both reproduced the V1→V2 effect (V2-316M – V1 $\Delta = +0.067$ [$+0.054$, $+0.084$] DoRothEA-only; $+0.056$ [$+0.041$, $+0.073$] CollecTRI). The within-V2 null persisted under both regulon sources. Selectivity audits using label permutation collapsed every L2 AUC to chance (median 0.500 at every scale, mean within ± 0.01 of 0.5), and the leakage audit (PROBE-20) confirmed zero donor overlap across folds for all three probes.

Taken together, the three probes paint a single coherent picture across scales (Figures 1–3): hub-TF identity is encoded already at V1 and saturates there; TF→target edges gain from the V1→V2 corpus and architecture refresh but not from the additional 212M parameters between V2-104M and V2-316M; and cell-type identity remains at-or-below a count-vector baseline at every scale, regardless of which transformer layer we read from.

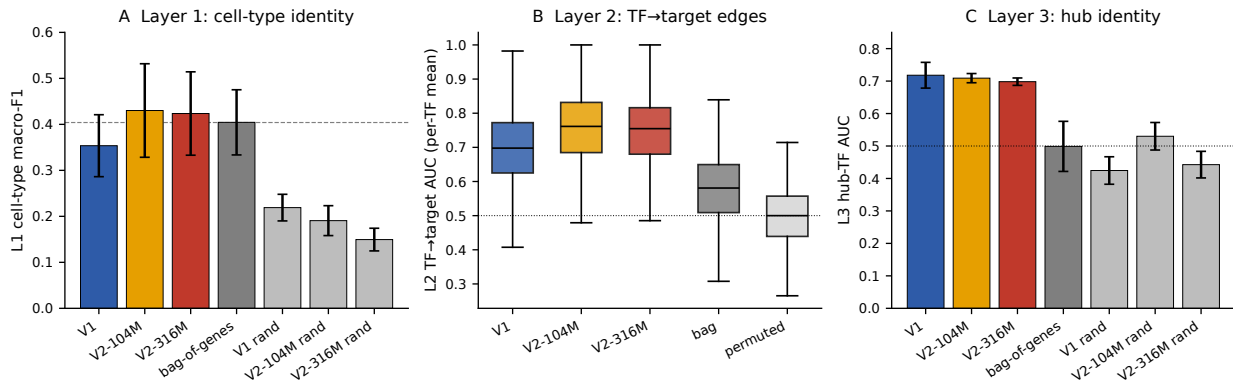


Figure 1: Three-scale headline numbers across all three probes. (A) Layer 1 (cell-type identity) macro-F1 across donor-stratified 5-fold CV; bars show mean across folds with 95% bootstrap CI. (B) Layer 2 (TF→target) per-TF AUC distribution; boxes show median and IQR over 375 TFs. (C) Layer 3 (hub identity) AUC across TF-stratified 5-fold CV. At every panel: trained Geneformer (V1, V2-104M, V2-316M); bag-of-genes baseline; random-init Geneformer at matched architecture/dimension.

4 Discussion

4.1 What we can conclude

Restricting attention to the only clean parameter-scaling axis the public Geneformer checkpoints offer — V2-104M → V2-316M, a $3\times$ increase in parameters at fixed pretraining corpus, fixed vocabulary, and (per the model cards) the same training schedule — none of our three probes show a measurable improvement. Cell-type identity moves by $\Delta = -0.007$ macro-F1 (CI [-0.034 , $+0.017$]); TF→target structure by $\Delta = -0.003$ AUC (CI [-0.014 , $+0.008$], $p=0.45$ over $n=375$ TFs); hub-TF identity by $\Delta = -0.011$ AUC (CI [-0.028 , $+0.007$]). The combined V1→V2 transition does deliver large gains on the L2 TF→target probe ($+0.054$ AUC, $p=10^{-16}$) and a modest, power-limited gain on L1 cell-type identity ($+0.07$ macro-F1, $n=5$, $p=0.063$), but those gains conflate three changes — parameter count, $\sim 3.5\times$ corpus expansion, and a vocabulary refresh — and we cannot attribute them to any one factor. The cleanest positive finding in the paper is L3: hub topology is encoded in the input gene-token embeddings of V1-10M at AUC 0.72, exceeds bag-of-

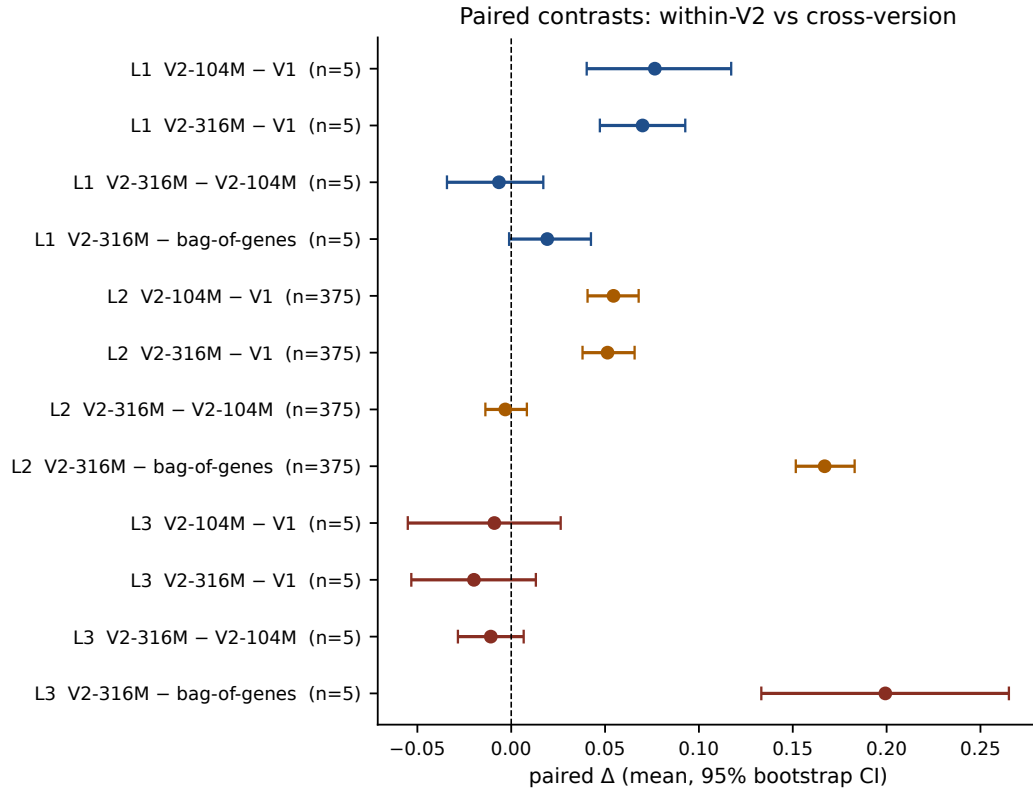


Figure 2: Forest plot: paired scaling deltas across V1→V2 (confounded) and within V2 (parameter-only). For each probe, three contrasts: V1→V2-104M, V1→V2-316M, and V2-104M→V2-316M. Within-V2 paired deltas are within noise of zero on every probe; the V1→V2 deltas on L1 and L2 are real but conflate parameter, corpus, and vocabulary changes. Whiskers are 95% bootstrap CIs.

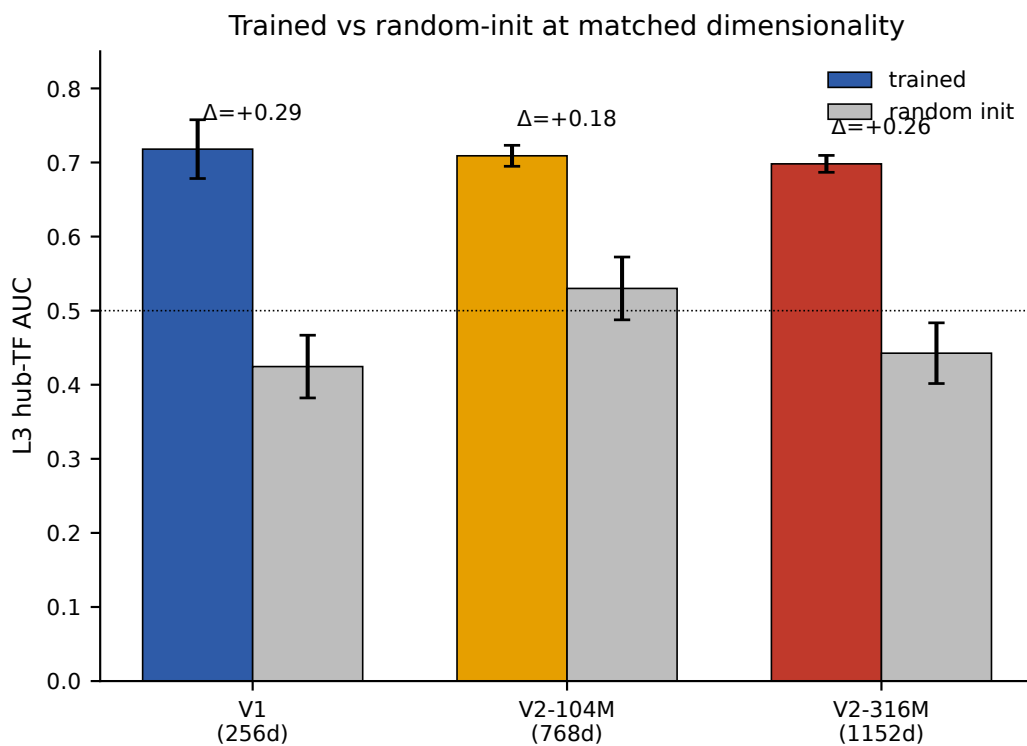


Figure 3: Geometry-falsifying random-init control across embedding dimensions. Trained vs. random-init L3 hub-AUC at 256-d (V1-10M), 768-d (V2-104M), and 1152-d (V2-316M). The trained-minus-random gap exceeds 0.18 AUC at every dimension, ruling out the alternative that hub identity is recoverable purely from geometric room.

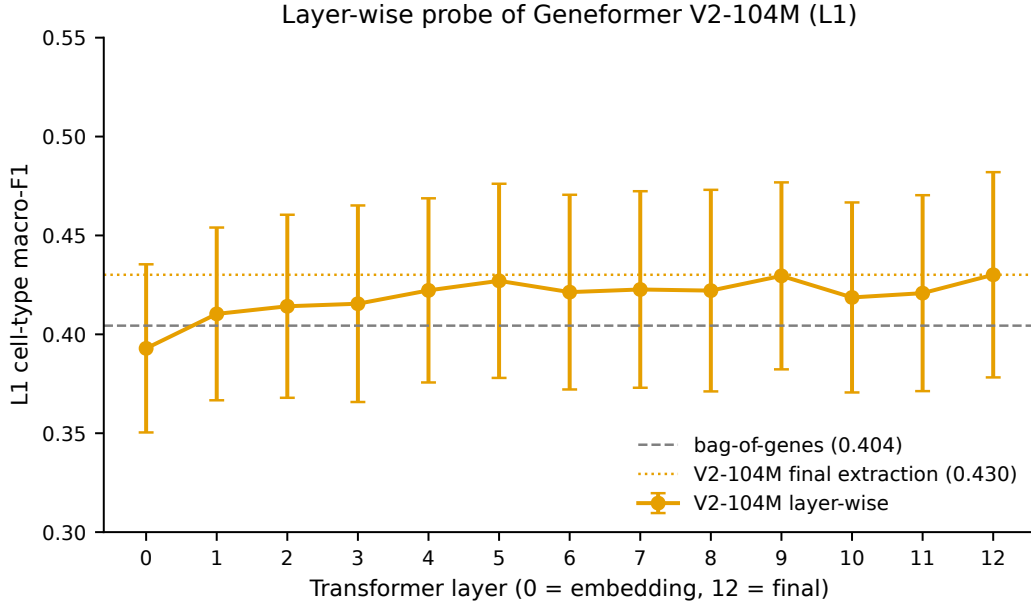


Figure 4: Layer-wise V2-104M Layer 1 macro-F1. Cell-type-identity macro-F1 of a fresh linear probe trained on the mean-pooled hidden state at each of the 13 layers (L0 = pre-transformer token embeddings; L1–L12 = transformer blocks). Solid line: mean across 5 donor-stratified folds. Shaded band: ± 1 SE. Dashed horizontal line: bag-of-genes baseline (0.404). No intermediate layer crosses the bag-of-genes line; the final layer used in the headline results (0.430) ties for best.

genes by +0.20 AUC at every scale, and the gap survives a random-initialization control at matched embedding dimensionality (random-init AUC 0.42 at 256d, 0.53 at 768d, 0.44 at 1152d). This rules out the alternative interpretation that 256-d (or 768-d, or 1152-d) Euclidean space simply has enough geometric room to separate any small subset of gene tokens from the rest. Hub identity is real model knowledge, it is encoded at the smallest variant, and it does not benefit measurably from $31\times$ more parameters or $\sim 3.5\times$ more pretraining data. Within Geneformer, hub identity is a low-complexity property — recoverable from co-expression statistics at modest model capacity — that saturates immediately.

4.2 What we cannot conclude

Several things we deliberately do not claim. We do not claim that parameter scaling fails to help scFMs in general; we tested one model family and one parameter-scaling step, and a null at this specific step does not generalize. We do not claim to have decomposed the V1→V2 gain into parameter-vs-data contributions; doing so cleanly would require checkpoints that hold all-but-one factor constant — a 10M-parameter Geneformer trained on the V2 corpus, or a 104M-parameter Geneformer trained on Genecorpus-30M — and neither exists in public release. We do not claim Geneformer “fails to encode cell-type identity”; the V1→V2 gain on L1 is real, if power-limited, and the L2 TF→target signal at V2-316M is large and selectivity-passing. We do not claim hub identity is generically a low-complexity signal across all single-cell foundation models; the claim is specifically that Geneformer encodes hub topology above bag-of-genes at every tested scale. Establishing the

same property for scGPT, UCE, scFoundation, or Nicheformer requires running the L3 probe on those embeddings, which we did not.

The within-V2 nulls on L1 and L3 are explicitly power-limited. With $n=5$ paired-fold observations the smallest two-sided Wilcoxon p attainable is 0.063 and our 95% bootstrap CI on the V2-104M $\rightarrow$ V2-316M difference rules out improvements larger than roughly +0.02 macro-F1 (L1) or +0.01 AUC (L3); smaller effects we cannot detect at this n . The L2 within-V2 null is on firmer footing — it pairs across $n=375$ TFs rather than 5 folds and has a CI that excludes anything larger than +0.008 AUC — but the L1 and L3 statements remain “no measurable improvement at our power” rather than “no improvement.” The L1 V2-316M-vs-bag-of-genes comparison ($\Delta=+0.019$, $p=0.31$) is similarly a failure to reject the null hypothesis of equality, not a positive demonstration of equivalence; we did not pre-specify an equivalence margin and did not perform two one-sided tests, so the careful phrasing throughout this paper is “fails to exceed” rather than “matches” or “is equivalent to.” Finally, our three probes are all linear logistic regression on frozen embeddings of healthy human blood/immune cells; the signal we find or fail to find on those probes is silent on what a non-linear probe, a different tissue, or a fine-tuned classifier would see.

4.3 Connection to prior work

Our finding that the largest publicly available Geneformer variant fails to exceed a bag-of-genes baseline on cell-type identity is consistent with the emerging consensus that scFMs do not reliably beat raw-expression baselines on tasks they were ostensibly designed to solve (Boiarsky et al., 2025). What we add to that consensus is a controlled three-scale parameter sweep on a single model family, run with the full Hewitt-Liang selectivity discipline plus a random-initialization control at matched dimensionality, plus an explicit decomposition of the V1 $\rightarrow$ V2 transition into parameter and corpus components. The selectivity-passing positive findings we do report (L3 hub identity at every scale, L2 TF $\rightarrow$ target at V2 scale) are stronger than the typical scFM benchmark precisely because of those controls.

The Theodoris-group scaling paper (Venkatesh et al., 2026) reports monotonic accuracy improvements with Geneformer scale on fine-tuned classifiers. Our null on within-V2 parameter scaling for frozen-embedding linear probes is not in conflict with that result. Fine-tuning lets the model rewrite its representation and frozen probing does not, and the two regimes can scale differently. The most direct reading consistent with both findings is that the Venkatesh et al. gains arise during fine-tuning rather than being available off-the-shelf in the pretrained representation — which is a meaningful distinction for any practitioner deciding whether to deploy V2-316M frozen embeddings as features in a downstream pipeline. Civale et al. (2026, arXiv preprint) recently showed that intermediate transformer layers can substantially outperform final-layer pooling for trajectory and perturbation tasks in scFoundation and Tahoe-X1. We use final-layer mean-pool (Reimers and Gurevych, 2019) throughout the headline numbers; layer-wise probing on V2-104M for our L1 cell-identity probe is robust to layer choice (no intermediate layer beats final-layer mean-pool on macro-F1). The Civale-style layer-wise gain may exist for our L2 or L3 probes — that is open.

4.4 What the field should take away

The most actionable observation from this study, for users of Geneformer’s published checkpoints, is that the apparent gain “from scale” between V1 and V2 is concentrated in the corpus-expansion and vocabulary-refresh axes, not in the parameter axis. Pure parameter scaling within the V2 family — $3\times$ more parameters at fixed data — does not measurably improve any of our three frozen-embedding probes. This is the single-cell analog of the Chinchilla observation (Hoffmann et al., 2022) that for compute-optimal training, parameter count and training tokens should scale jointly, and that scaling parameters at fixed data is the wrong axis to push. Practitioners selecting between V2-104M and V2-316M as a frozen feature extractor should expect roughly equivalent performance on the kinds of probes we ran, and should weigh the $3\times$ inference-cost difference accordingly. Practitioners considering whether to push for a hypothetical V3 with more parameters at the same $\sim 104\text{M}$ -cell corpus should expect, by the same logic, diminishing returns on frozen embeddings; the more productive next axis is data, not parameters. Our results say nothing about the parameter-vs-data trade-off for fine-tuned downstream classifiers, where the Venkatesh et al. (2026) result is the more relevant precedent.

A second, more general takeaway concerns probing methodology. The selectivity control (Hewitt and Liang, 2019) and the random-initialization control (Zhang and Bowman, 2018) cost almost nothing to run — a re-execution of the same probe with permuted labels and an untrained-but-same-architecture model — and they substantively change what one is allowed to claim. Random initialization at V1’s 256-d alone produces an L3 hub-AUC of 0.42 and at V2-316M’s 1152-d an AUC of 0.44; without the matched-dimensionality random-init control, an unsuspecting reader could read our L3 result as “linear separability of any small class in 1000-d Euclidean space” rather than as model knowledge. We urge that future scFM benchmarking adopt the selectivity-plus-random-init pair as a default, not an optional add-on.

4.5 Future work

Four extensions seem most useful. First, systematic layer-wise probing of Geneformer for the L2 TF→target and L3 hub-identity probes — the Civale et al. (2026) result that intermediate layers carry task-relevant signal that final-layer pooling discards is established for other scFMs and other tasks, and our L1 layer-wise null does not generalize automatically to gene-relational probes. Second, replication of all three probes on scGPT, UCE, and scFoundation. The L3 hub claim in particular needs cross-model evidence before it can be promoted from “Geneformer encodes hub topology” to “scFMs encode hub topology.” Third, cross-tissue generalization. Healthy human blood/immune is a single tissue with strong cell-type structure and a deep regulatory literature; whether the same probes report the same scaling patterns on brain, gut, or tumor tissue is unknown. Fourth — and this is a request to model trainers more than a research direction — a Geneformer checkpoint suite that holds all-but-one factor constant (a 10M-parameter Geneformer on the V2 corpus, or a 104M-parameter Geneformer on Genecorpus-30M) would let the field cleanly decompose the V1→V2 confound that we and the Theodoris group are currently both unable to resolve. Until such a suite exists, scaling claims for scFMs that compare across model-family transitions

will continue to confound parameter count with corpus and vocabulary change, and any single-axis interpretation of those comparisons will rest on assumption rather than evidence.

5 Reproducibility

Code, environment, and data-pull scripts are committed at <https://github.com/williamcolegithub/Morpheus> (made public concurrent with this preprint release) and archived at Zenodo, concept DOI [10.5281/zenodo.20046043](https://doi.org/10.5281/zenodo.20046043) (always resolves to the latest version; the version snapshot accompanying this preprint is v1.0-preprint, DOI [10.5281/zenodo.20046044](https://doi.org/10.5281/zenodo.20046044), 6 May 2026). Pinned uv.lock; deterministic seed 0 threaded through every random component. Census release 2025-11-08; decoupler 2.1.6; transformers 4.57; PyTorch 2.11.0+mps (laptop, current PyPI release at extraction time) and PyTorch 2.5.1+cu124 (AWS EC2; the cloud CUDA 12.4 driver predates the cu130-built 2.11 wheels, so we pinned the cloud env to 2.5.1+cu124 — the only library version that differs across laptop and cloud, and it does not affect any numerical result). Pipeline reproduces end-to-end on a 24 GB MacBook Air (V1) plus two AWS EC2 g5.xlarge sessions (V2-104M and V2-316M, ~\$5 each). Full per-fold and per-TF results are released as parquet files; figures are regenerable via `morpheus.analysis.figures_v2`.

Data availability

All data used in this work are publicly available from existing repositories. The single-cell expression corpus is the human blood/immune subset of the CZ CELLxGENE Census release 2025-11-08, accessed via the `cellxgene_census` Python package (CZI Cell Science Program et al., 2025). Transcription-factor regulons are DoRothEA confidence levels A/B/C (Garcia-Alonso et al., 2019) and CollecTRI (Müller-Dott et al., 2023), both accessed via decoupler (Badia-i Mompel et al., 2022). Pretrained Geneformer model weights and gene-vocabulary dictionaries are distributed at <https://huggingface.co/ctheodoris/Geneformer> (Theodoris et al., 2023). The exact data-pull scripts used to fetch and preprocess these resources are committed in the project repository (see Code availability).

Code availability

The full reproducible pipeline — data acquisition, embedding extraction, probe runners, sensitivity sweep, statistics, and figure generation — is committed at <https://github.com/williamcolegithub/Morpheus> under the MIT License, and archived at Zenodo with concept DOI [10.5281/zenodo.20046043](https://doi.org/10.5281/zenodo.20046043) (the v1.0-preprint snapshot accompanying this manuscript is at DOI [10.5281/zenodo.20046044](https://doi.org/10.5281/zenodo.20046044)). All per-fold and per-TF results, the stats summary, the bibliography, and the figure-generation source are included.

Acknowledgments

The author thanks the developers of the open-source resources this work depends on: the CZ CEL-LxGENE team for the Census infrastructure; the Saez-Rodriguez group for DoRothEA, CollecTRI, and decoupler; the Theodoris group for releasing Geneformer weights, vocabulary dictionaries, and rank-value tokenization code; and the maintainers of scanpy, anndata, scikit-learn, transformers, and PyTorch. Compute for V2-104M and V2-316M extractions was self-funded on AWS EC2.

Competing interests

The author declares no competing interests.

Funding

This work received no external funding. Compute and infrastructure costs were self-funded.

References

- Tamim Abdelaal, Lieke Michielsen, Davy Cats, Dyon Hoogduin, Hailiang Mei, Marcel J. T. Reinders, and Ahmed Mahfouz. A comparison of automatic cell identification methods for single-cell RNA sequencing data. *Genome Biology*, 20:194, 2019.
- Constantin Ahlmann-Eltze and Wolfgang Huber. Comparison of transformations for single-cell RNA-seq data. *Nature Methods*, 20:665–672, 2023.
- Constantin Ahlmann-Eltze, Wolfgang Huber, and Simon Anders. Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines. *Nature Methods*, 22:1657–1661, 2025.
- Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes, 2016. arXiv:1610.01644 (preprint).
- Pau Badia-i Mompel, Jesús Vélez Santiago, Jana Braunger, Celina Geiss, Daniel Dimitrov, Sophia Müller-Dott, Petr Taus, Aurelien Dugourd, Christian H. Holland, Ricardo O. Ramirez Flores, and Julio Saez-Rodriguez. decoupler: ensemble of computational methods to infer biological activities from omics data. *Bioinformatics Advances*, 2(1):vbac016, 2022.
- Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219, 2022.
- Rebecca Boiarsky, Nalini M. Singh, Andrés Buendía, Ava P. Amini, Caroline Uhler, and Christina V. Theodoris. Biology-driven insights into the power of single-cell foundation models. *Genome Biology*, 26:167, 2025.

- Vincenzo Yuto Civalé, Roberto Semeraro, Andrew David Bagdanov, and Alberto Magi. Intermediate layers encode optimal biological representations in single-cell foundation models, 2026. arXiv preprint arXiv:2604.14838.
- Alexis Conneau, German Kruszewski, Guillaume Lample, Loic Barrault, and Marco Baroni. What you can cram into a single vector: Probing sentence embeddings for linguistic properties. In *Proceedings of ACL 2018*, 2018.
- Haotian Cui, Chloe Wang, Hassaan Maan, Kuan Pang, Fengning Luo, Nan Duan, and Bo Wang. scgpt: toward building a foundation model for single-cell multi-omics using generative ai. *Nature Methods*, 21:1470–1480, 2024. doi: 10.1038/s41592-024-02201-0.
- CZI Cell Science Program, Shibli Abdulla, Brian Aevermann, Pedro Assis, et al. CZ CELLxGENE discover: a single-cell data platform for scalable exploration, analysis and modeling of aggregated data. *Nucleic Acids Research*, 53:D886–D900, 2025.
- Alexander D. Diehl, Terrence F. Meehan, Yvonne M. Bradford, Matthew H. Brush, Wasila M. Dahdul, David S. Dougall, Yongqun He, David Osumi-Sutherland, Alan Ruttenberg, Sirarat Sarntivijai, Ceri E. Van Slyke, Nicole A. Vasilevsky, Melissa A. Haendel, Judith A. Blake, and Christopher J. Mungall. The cell ontology 2016: enhanced content, modularization, and ontology interoperability. *Journal of Biomedical Semantics*, 7:44, 2016.
- Bradley Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7: 1–26, 1979.
- Luz Garcia-Alonso, Christian H. Holland, Mahmoud M. Ibrahim, Dénes Türei, and Julio Saez-Rodriguez. Benchmark and integration of resources for the estimation of human transcription factor activities. *Genome Research*, 29:1363–1375, 2019.
- Minsheng Hao, Jing Gong, Xin Zeng, Chiming Liu, Yucheng Guo, Xingyi Cheng, Taifeng Wang, Jianzhu Ma, Xuegong Zhang, and Le Song. Large-scale foundation model on single-cell transcriptomics. *Nature Methods*, 21:1481–1491, 2024. doi: 10.1038/s41592-024-02305-7.
- John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Proceedings of EMNLP-IJCNLP 2019*, 2019.
- Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, et al. Training compute-optimal large language models. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*, 2022.
- Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models, 2020. arXiv:2001.08361.
- Kasia Z. Kedzierska, Lorin Crawford, Ava P. Amini, et al. Zero-shot evaluation reveals limitations of single-cell foundation models. *Genome Biology*, 26:101, 2025.

- Nikolaus Kriegeskorte, Marieke Mur, and Peter Bandettini. Representational similarity analysis – connecting the branches of systems neuroscience. *Frontiers in Systems Neuroscience*, 2:4, 2008.
- Sophia Müller-Dott, Eirini Tsirvouli, Miguel Vazquez, Ricardo O. Ramirez Flores, Pau Badia-i Mompel, Robin Fallegger, Dénes Türei, Astrid Læg Reid, and Julio Saez-Rodriguez. Expanding the coverage of regulons from high-confidence prior knowledge for accurate estimation of transcription factor activities. *Nucleic Acids Research*, 51:10934–10949, 2023.
- Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, et al. Pytorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems 32* (NeurIPS 2019), 2019.
- Aditya Pratapa, Amogh P. Jalihal, Jeffrey N. Law, Aditya Bharadwaj, and T. M. Murali. Benchmarking algorithms for gene regulatory network inference from single-cell transcriptomic data. *Nature Methods*, 17:147–154, 2020.
- Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of EMNLP-IJCNLP 2019*, 2019.
- Anna C. Schaar, Alejandro Tejada-Lapueta, Giovanni Palla, Robert Gutgesell, Lennard Halle, Mariia Minaeva, Larsen Vornholz, Leander Dony, Francesca Drummer, Mojtaba Bahrani, and Fabian J. Theis. Nicheformer: a foundation model for single-cell and spatial omics. *Nature Methods*, 2025. doi: 10.1038/s41592-025-02814-z.
- Christina V. Theodoris, Ling Xiao, Anant Chopra, Mark D. Chaffin, Zeina R. Al Sayed, Matthew C. Hill, Helene Mantineo, Elizabeth M. Brydon, Zexian Zeng, X. Shirley Liu, and Patrick T. Ellinor. Transfer learning enables predictions in network biology. *Nature*, 618:616–624, 2023. doi: 10.1038/s41586-023-06139-9.
- Madhavan S. Venkatesh, Siddharth V. Mahesh, Tarak N. Nandi, Ravi K. Madduri, Karin Pelka, and Christina V. Theodoris. Scaling and quantization of large-scale foundation model enables resource-efficient predictions in network biology. *Nature Computational Science*, 2026.
- Frank Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1:80–83, 1945.
- F. Alexander Wolf, Philipp Angerer, and Fabian J. Theis. SCANPY: large-scale single-cell gene expression data analysis. *Genome Biology*, 19:15, 2018.
- Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Remi Louf, Morgan Funtowicz, et al. Transformers: State-of-the-art natural language processing. In *Proceedings of EMNLP 2020: System Demonstrations*, pages 38–45, 2020.
- Kelly W. Zhang and Samuel R. Bowman. Language modeling teaches you more than translation does: Lessons learned through auxiliary task analysis. In *Proceedings of BlackboxNLP at EMNLP 2018*, 2018.

Kip D. Zimmerman, Mark A. Espeland, and Carl D. Langefeld. A practical solution to pseudoreplication bias in single-cell studies. *Nature Communications*, 12:738, 2021.